

MA 301
Test 1
Fall - 2014

Problem	points	out of
1		7
2		7
3		7
4		4
5		7
6		8
7		2
EC		5
Score		42

Name: SOLUTIONS

HR: _____

Signature: _____

READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Let $F = [z^2, x - y, 7]$ and C be the intersection of the surfaces $x = y + z$ and $y = z^2$ with $0 \leq y \leq 9$.

(a) Compute $\int_C F \cdot dr$

$$\vec{r}(t) = [t^{1/2} + t, t, t^{1/2}] \quad 0 \leq t \leq 9$$

$$r'(t) = [\frac{1}{2}t^{-1/2} + 1, 1, \frac{1}{2}t^{-1/2}]$$

$$\vec{F}(\vec{r}(t)) = [t, t^{1/2} + t - t, 7] = [t, t^{1/2}, 7]$$

$$= \int_0^9 \left(\frac{1}{2}t^{1/2} + t + t^{1/2} + \frac{7}{2}t^{-1/2} \right) dt$$

$$= \int_0^9 \left(\frac{3}{2}t^{1/2} + t + \frac{7}{2}t^{-1/2} \right) dt$$

$$= \left[t^{3/2} + \frac{1}{2}t^2 + 7t^{1/2} \right]_0^9$$

$$= 9^{3/2} + \frac{1}{2}9^2 + 7 \cdot 9^{1/2} - 0$$

$$= 27 + \frac{81}{2} + 21 = \boxed{\frac{177}{2} \text{ or } 88.5}$$

(b) Completely set-up, but DO NOT compute the integral needed to find the length of C . (arclength = $\int_C 1 ds$)

Note $ds = |r'(t)| dt$

$$\text{and } |r'(t)| = \sqrt{\left(\frac{1}{2}t^{-1/2} + 1\right)^2 + 1 + \left(\frac{1}{2}t^{-1/2}\right)^2}$$

$$\Rightarrow \boxed{\begin{array}{l} \text{Length} \\ \text{of} \\ C \end{array} = \int_0^9 \sqrt{\left(\frac{1}{2}t^{-1/2} + 1\right)^2 + 1 + \frac{1}{4}t^{-1}} dt}$$

Problem 2: Let $\mathbf{F} = \left[y - \ln(y), -\frac{x}{y} + x, 1 \right]$ compute the work required to move an object along C (shown below). Justify your work.

Want f such that $\vec{F} = \nabla f$

$$f_x = y - \ln y$$

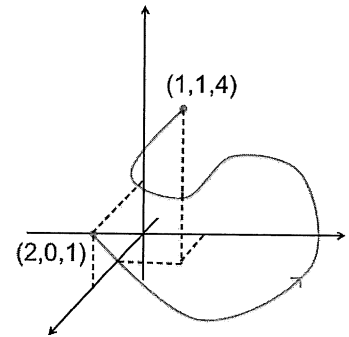
$$f = xy - x \ln y + g(y, z)$$

$$-\frac{x}{y} + x = f_y = x - \frac{x}{y} + g_y \Rightarrow g_y = 0 \Rightarrow g = h(z)$$

$$f = xy - x \ln y + h(z)$$

$$1 = f_z = 0 + h_z \Rightarrow h_z = 1 \Rightarrow h_z = z + C$$

$$\boxed{f = xy - x \ln y + z} \quad (C=0)$$



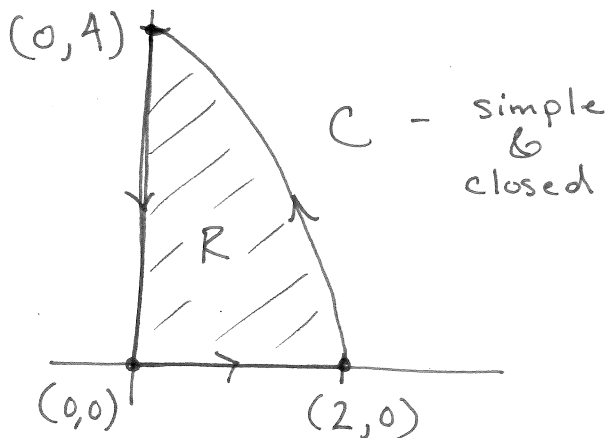
Problem 3: Let $F(x,y) = \overbrace{(xe^y - 3xy)\mathbf{i}}^M + \overbrace{\frac{3}{2}y^2\mathbf{j}}^N$ and suppose C consist of the following 3 curves: the line segment from $(0,0)$ to $(2,0)$ then the curve $y = 4 - x^2$ from $(2,0)$ to $(0,4)$ and the line segment from $(0,4)$ to $(0,0)$.

(a) Sketch C

(b) Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

$$N_x = 0$$

$$M_y = xe^y - 3x$$



$$= \iint_R 0 - xe^y + 3x \, dA$$

$$= \int_0^2 \int_0^{4-x^2} (-xe^y + 3x) \, dy \, dx$$

$$= \int_0^2 -xe^y + 3xy \Big|_0^{4-x^2} \, dx$$

$$= \int_0^2 -xe^{4-x^2} + 3x(4-x^2) \, dx$$

$$= \int_0^2 -xe^{4-x^2} + 12x - 3x^3 \, dx$$

$$= \int_{x=0}^{x=2} \frac{1}{2}e^u \, du + \left(6x^2 - \frac{3}{4}x^4 \right) \Big|_0^2$$

$$= \frac{1}{2}e^{4-x^2} + 6x^2 - \frac{3}{4}x^4 \Big|_0^2$$

$$= \left(\frac{1}{2}e^0 + 24 - 12 \right) - \left(\frac{1}{2}e^4 \right)$$

$$= \frac{1}{2} + 12 - \frac{1}{2}e^4$$

$$= \boxed{\frac{25 - e^4}{2} \quad \text{or} \quad -14.8}$$

$$u = 4 - x^2$$

$$du = -2x \, dx$$

Problem 4: Suppose a straight wire from initial point $\overset{P}{(0,1,5)}$ to end point $\overset{Q}{(2,2,3)}$ has a linear density function given by $\delta(x,y,z) = x^3z$. Find the total mass of the wire.

$$\text{Mass} = (\text{Linear density}) (\text{Length})$$

$$dM = \delta \, ds$$

$$\Rightarrow \text{Mass} = \int_c^b dM = \int_c^b \delta \, ds = \int_a^b \delta(\vec{r}(t)) |\vec{r}'(t)| \, dt$$

$$\vec{r}(t) = (Q-P)t + P = [2t, t+1, -2t+5] \quad 0 \leq t \leq 1$$

$$\vec{r}'(t) = [2, 1, -2]$$

$$|\vec{r}'(t)| = \sqrt{4+1+4} = \sqrt{9} = 3$$

$$\delta(\vec{r}(t)) = (2t)^3(-2t+5) = 8t^3(-2t+5) = -16t^4 + 40t^3$$

$$\Rightarrow \text{Mass} = \int_0^1 (-16t^4 + 40t^3) (3) \, dt$$

$$= \left. -\frac{48}{5}t^5 + 30t^4 \right|_0^1$$

$$= -\frac{48}{5} + 30$$

$$= \boxed{\frac{102}{5} \quad \text{or} \quad 20.4}$$

Problem 5: Answer the following true/false questions

- (a) True or False: If C is an ellipse and F is irrotational then $\int_C F \cdot dr = 0$
closed conservative
- (b) True or False: If $\int_C F(r) \cdot dr \neq 0$ for some closed path C , then F is conservative.
- (c) True or False: If $F = \nabla f$ then F is conservative.
- (d) True or False: If F is conservative then the work required to move an object along any path is zero.
- (e) True or False: A vector field, F , can be both incompressible and irrotational.
- (f) True or False: If F is path-independent and C is any closed curve, then $\int_C F(r) \cdot dr = 0$
- (g) True or False: Line integrals over all constant vector fields are path-independent.

Problem 6: Let F be the vector field and C_1 and C_2 be the curves shown in the provided figure.

- (a) Is F incompressible? No
- (b) Is F irrotational? Yes
- (c) Is F conservative? Yes
- (d) Would the work for a mass traveling along C_1 be positive, negative, or zero? Explain.

Negative; the movement acts against the force along most of the path
 ($F \cdot dr$ is negative on most of C_1)

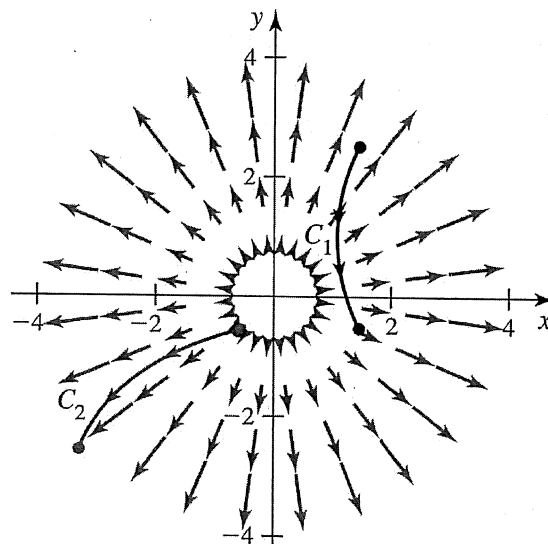
- (e) Would the work for a mass traveling along C_2 be positive, negative, or zero? Explain.

Positive; the movement acts in the same direction at every point of C_2
 ($F \cdot dr$ is always positive)

- (f) Which is larger: $|\text{curl}(F(2,2))|$ or $|\text{div}(F(2,2))|$ Explain.

$\text{div}(\vec{F}(2,2))$ is positive since the outflow around $(2,2)$ is greater than the inflow. $\therefore \text{div}(\vec{F}(2,2)) > \text{curl}(\vec{F}(2,2))$

$\text{curl}(\vec{F}(2,2))$ is zero since $\vec{F}$ is conservative.



Problem 7: Let $f(x,y,z)$ be a scalar-valued function. Using the concepts from this class, explain why $\text{curl}(\nabla f) = 0$ no matter what $f(x,y,z)$ is. If you cannot explain it, then prove it mathematically.

∇f always creates a conservative vector field. Therefore,
 $\text{curl}(\nabla f) = \vec{0}$

To show this mathematically, assume f has continuous partials
 and $\text{curl}([f_x, f_y, f_z]) = [f_{zy} - f_{yz}, f_{xz} - f_{zx}, f_{yx} - f_{xy}]$
 $= [0, 0, 0]$

Extra Credit: Let C be a simple closed curve enclosing a region R .

(a) Use Green's Theorem to show $\oint_C x dy = \text{Area}(R)$

$$\oint_C x dy \hookrightarrow \oint_C \vec{F} \cdot d\vec{r} \quad \text{where} \quad \vec{F} = [0, x]$$

$$\Rightarrow N_x = 1$$

$$M_y = 0$$

$$\oint_C x dy = \iint_R (1 - 0) dA = \iint_R dA$$

$$= \text{Area}(R)$$

(b) Use this to find the area of the region with boundary given by $\mathbf{r}(t) = [\sin 2t, 2 \sin t]$ with $0 \leq t \leq \pi$.

Hint: The trig identity: $\sin(2t) = 2 \sin t \cos t$ might come in handy.

From above:

$$\begin{aligned} \text{Area}(R) &= \oint_C \vec{F} \cdot d\vec{r} \\ &= \int_0^\pi \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \\ &= \int_0^\pi 2 \cos t \sin 2t dt \\ &= \int_0^\pi 2 \cos t (2 \sin t \cos t) dt \\ &= 4 \int_0^\pi \sin t \cos^2 t dt \\ &= -4 \int_{t=0}^{t=\pi} u^2 du \\ &= -4 \left. \frac{u^3}{3} \right|_{t=0}^{t=\pi} = -\frac{4}{3} (\cos t)^3 \Big|_0^\pi = -\frac{4}{3} [(\cos(\pi))^3 - (\cos(0))^3] \\ &= \boxed{\frac{8}{3}} \end{aligned}$$

where $\vec{F} = [0, x]$

$$\vec{r}(t) = [\sin 2t, 2 \sin t]$$

$$\vec{r}'(t) = [2 \cos 2t, 2 \cos t]$$

$$0 \leq t \leq \pi$$

$$\vec{F}(\vec{r}(t)) = [0, \sin 2t]$$

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = 2 \cos t \sin 2t$$

$$u = \cos t$$

$$du = -\sin t dt$$

